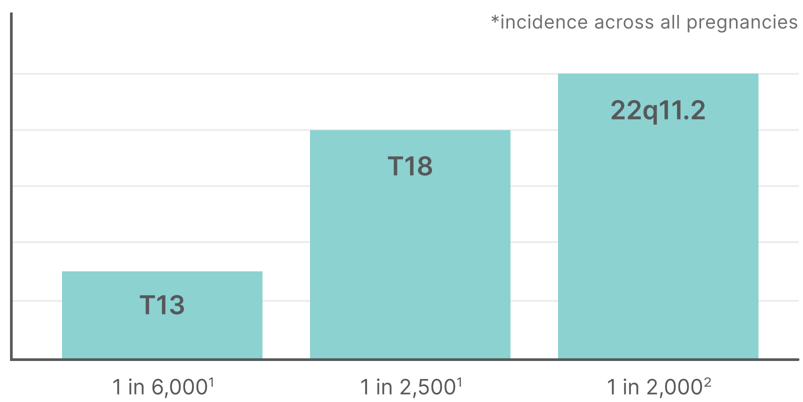
UNITY
by BILLIONTOONE

UNITY Aneuploidy NIPT: 22q11.2 Microdeletion Analysis

For all singleton and twin pregnancies

22q11.2 Deletion Syndrome is **Prevalent**

22q11.2 (also known as DiGeorge Syndrome) is more prevalent than commonly screened for chromosomal conditions.



Early **Identification** is Difficult with Ultrasound

Ultrasound findings are often not unique to 22q11.2 and can make diagnosis difficult.

90% of 22q11.2 microdeletions are de-novo and prevalence does not increase with maternal age.[3]

Early Prenatal **Diagnosis** Can Make a Difference

Prenatal detection of 22q11.2 Deletion Syndrome is associated with less cardiac and noncardiac morbidity.[4]

Some babies can develop neonatal hypocalcemia that can cause seizures if not treated.[5]

Expertise in Fetal DNA Quantification Allows for **Highly Accurate** Detection of 22q11.2 Microdeletion

Includes the full A-D region AND nested microdeletions.[6]

>95%
sensitivity

>99.9%
specificity

80%
PPV

Fetal DNA Quantification Overcomes the Technical Limitations with SNP Based Tests

QCT Counting Technology Allows for Extreme **Precision and Sensitivity**

We are experts at fetal DNA quantification, which allows for detecting small changes, even down to a single base pair.

QCT technology provides increased accuracy for smaller chromosomal changes, like 22q11.2.

This makes it possible to provide industry-leading sensitivity and PPV.

Aneuploidy
Easiest to detect

Microdeletion
More difficult to detect

Recessive conditions
Requires extreme precision and sensitivity

- Cystic fibrosis
- Spinal muscular atrophy
- Hemoglobinopathies

10+ million base pairs — **1+ million** base pairs — **Single** base pairs

UNITY Aneuploidy Screen. Backed by clinical data.

New Publication: *Performance Characteristics of a Next Generation Sequencing-Based cfDNA Assay for Common Aneuploidies in a General Risk Population*[7]

Mean test characteristics n=114,707

MATERNAL AGE
29 years old

GESTATIONAL AGE
13.9 weeks

FETAL FRACTION
9.30% (1.5-39%*)

TURNAROUND TIME
5 days

	Trisomy 21	Trisomy 18	Trisomy 13	Combined Autosomes
Sensitivity	99.7%	99.5%	>99.9%	99.7%
Specificity	99.7%	>99.9%	>99.9%	99.9%
PPV	90.5%	97.6%	73.3%	90.8%
NPV	>99.9%	>99.9%	>99.9%	>99.9%

*1.5-39% represents the full distribution of fetal fraction

Access the peer-reviewed publication here

Also available with UNITY Aneuploidy Screen

Add-on at any point during the patient's pregnancy

+ UNITY Fetal RhD NIPT

+ UNITY Fetal Antigen NIPT

Twin zygosity determination*

*included with every twin order

650.460.2551 | support@unityscreen.com | unityscreen.com

1. Goel, Nitin et al. "Trisomy 13 and 18-Prevalence and mortality-A multi-registry population based analysis." American journal of medical genetics. Part A vol. 179,12 (2019): 2382-2392. doi:10.1002/ajmg.a.61365. 2. Fung, W., Butcher, N., Costain, G. et al. Practical guidelines for managing adults with 22q11.2 deletion syndrome. Genet Med 17, 599–609 (2015). https://doi.org/10.1038/gim.2014.175. 3. McDonald-McGinn DM, Hain HS, Emanuel BS, et al. 22q11.2 Deletion Syndrome. 1999 Sep 23 [Updated 2024 May 9]. In: Adam MP, Feldman J, Mirzaa GM, et al., editors. GeneReviews® [Internet]. Seattle (WA): University of Washington, Seattle; 1993-2024. Available from: https://www.ncbi.nlm.nih.gov/books/NBK1523/. 4. Freud LR, et al. Prenatal vs postnatal diagnosis of 22q11.2 deletion syndrome: cardiac and noncardiac outcomes through 1 year of age. Am J Obstet Gynecol. 2024 Mar;230(3):368.e1-368.e12. doi: 10.1016/j.ajog.2023.09.005. Epub 2023 Sep 16. PMID: 37717890. 5. Cheung, E. N., George, S. R., Andrade, D. M., Chow, E. W. C., Silversides, C. K., & Bassett, A. S. (2014). Neonatal hypocalcemia, neonatal seizures, and intellectual disability in 22q11.2 deletion syndrome. Genetics in Medicine, 16(1), 40–44. https://doi.org/10.1038/gim.2013.71. 6. UNITY Aneuploidy NIPT Validation. Data on File. March 2024. 7. Wynn J, Rego S, Fang J, Alford B, Carter R, Hoskovec J. Performance Characteristics of a Next Generation Sequencing-Based cfDNA Assay for Common Aneuploidies in a General Risk Population. Ann Gynecol Obstetr Res. 2024; 7(1): 1027.

UN_FL_22Q_11-2_2407